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# Probe Directions Are Not Steering Knobs: Activation Evidence for an Agency-Preserving Social Cohesion Manifold

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**Abstract.** We study whether a language model's internal representations contain separable structure for agency-preserving social cohesion: cooperative relation that preserves truth, consent, dissent, privacy, appeal, and exit rights. The work is deliberately compute-first. It does not claim human behavioral or neural effects. We build deterministic paired benchmarks that contrast flexible contextual relation with pseudo-cohesion failures such as social-debt pressure, dissent suppression, verification blocking, privacy bypass, rigid us/them partitioning, and coercive unity language. Across the strongest current cue-balanced boundary-prior suite, the simple cue counter solves 0/36 pairs, while Qwen2.5 0.5B and 1.5B activation directions reach 1.000 leave-one-pair-out accuracy across five model/layer settings. Geometry audits reject an orthogonal-axis interpretation: mechanism directions are moderately aligned, and residual audits support a shared signed genuine-vs-pseudo manifold plus mechanism-specific residual subspaces. Causal steering experiments then expose the key bottleneck. Forward hooks move target hidden states almost exactly by the requested signed displacement and shift post-hook projections by roughly 11 units, but generated text scores move only 0.015 to 0.024. The main claim is therefore methodological: probe directions can be representation-real without being reliable generation-time controls. Publishable progress now depends on generated paraphrase hardening, anti-compliance controls, and monotonic steering protocols that jointly move hook telemetry, generated-output projections, and behavior.

## 1. Introduction

Generative models can increasingly produce dialogue, narratives, plans, and social explanations that feel conciliatory or prosocial. The obvious optimization target is engagement or agreement. The target here is stricter: can we monitor or steer outputs toward social cohesion without collapsing into sycophancy, dependence, coercion, or dissent suppression?

We define agency-preserving social cohesion as the capacity of persons and groups to remain in truthful, dignity-preserving, agency-respecting relation while pursuing shared goods or principled non-agreement under conflict. The definition is intentionally negative as well as positive. A system should not receive credit for "harmony" if it hides evidence, pressures refusal, blocks verification, erases privacy, or treats unity as a reason to remove exit rights.

This paper reports a compact version of the current repo state. The empirical evidence is synthetic and compute-only. The contribution is a research scaffold and a set of disciplined claims: (i) cue-balanced deterministic benchmarks can remove the easiest lexical shortcuts, (ii) signed activation directions separate agency-preserving from pseudo-cohesive examples across several Qwen model/layer points, (iii) the geometry is not near-orthogonal persona axes, and (iv) naive activation addition does not yet yield reliable semantic control at generation time.

**Current best claim.** The project has evidence for a signed activation manifold that separates controlled examples of flexible contextual relation from pseudo-cohesion, and evidence for a control bottleneck between hidden-state displacement and generated behavior. It does not yet have a validated human-effect claim or a clean causal steering claim.

## 2. Related Work and Conceptual Frame

The mechanistic premise is inspired by sparse autoencoder and activation steering work, where directions or features have been used to study refusal, sycophancy, persona traits, and other high-level model behaviors [1,2,3,4,5]. These results make it plausible to ask whether social-reasoning constructs have representational structure. They also warn against overuse of single-vector stories: low-dimensional probes can be brittle, entangled with style, and unreliable as causal controls.

The social premise is older. Repeated games and cooperative-AI work show that reciprocity, clarity, reputation, forgiveness, and institutional context matter for cooperation [6,7]. Social psychology adds that group identity, threat, status, and intergroup boundaries shape cooperation beyond payoff matrices. The project's boundary-prior benchmark operationalizes this insight: social breakdown can arise from rigid boundary reification ("they are outside the circle of evidence") or from coercive boundary collapse ("we are one, so refusal is betrayal").

The normative frame draws from pluralistic alignment and from *Magnifica Humanitas*, which emphasizes that technology is not neutral in practice because it carries the aims and power relations of the people and institutions that build and deploy it [8]. In this frame, the target is not a single harmony scalar. It is a constrained family of social capacities: repair, reciprocity, truthfulness, autonomy, constructive dissent, manipulation resistance, and privacy or exit rights.

## 3. Task Definition

Each benchmark example is a paired contrast. The positive side preserves a social relation while maintaining agency constraints. The negative side is a pseudo-cohesion failure: it uses cooperative, warm, loyal, safety-coded, or unity-coded language while removing one of the constraints.

Let  $\phi_l(x)$  be a model representation at layer  $l$ . For a paired dataset  $D = \{(x_{i+}, x_{i-})\}$ , the simplest contrastive direction is:

```
v_l = mean_i phi_l(x_i+) - mean_i phi_l(x_i-)
s_l(x) = dot(normalize(phi_l(x)), normalize(v_l))
```

The scientific target is not "maximize  $s_l$ " in isolation. A useful system must satisfy guardrails:

```
maximize repair + reciprocity + calibrated trust
subject to truth >= tau_truth
          autonomy >= tau_autonomy
          dissent_safety >= tau_dissent
          privacy >= tau_privacy
          exit_rights >= tau_exit
          manipulation_risk <= tau_manipulation
```

This distinction matters for evaluation. Agreement, warmth, and narrative force are not success metrics unless the constraints survive.

## 4. Methods

### 4.1 Datasets

The repo contains several deterministic local benchmark families. The early scripted scenario suite covers 25 social dilemmas and 450 scripted runs, but lexical and metrics-only baselines solve it, so it is retained as a pipeline sanity check. The stronger lanes are pseudo-cohesion, structural autonomy, and boundary-prior contrasts.

The current strongest set is the controlled cue-balanced boundary-prior expansion: 36 matched pairs and 72 activation prompts. It crosses six mechanisms with two negative poles, then wraps each contrast in three neutral record genres. The positive side is flexible contextual relation. The negative poles are rigid boundary reification and coercive boundary collapse.

#### 4.2 Leakage, Component, and Geometry Gates

Every pairwise dataset should pass a leakage gate before activation claims are trusted. The gate counts simple positive-minus-negative lexical cues and records cue-solved, tied, and inverted pairs. Passing this gate is not enough to prove semantics, but failing it means the benchmark is still surface-solvable.

We also run reviewer-style geometry diagnostics. Signed cosine alone can hide cancellation, and cosine near -1 is not independence. Reports therefore include signed and absolute off-diagonal cosine, anti-aligned-pair counts, residual pair-difference energy after the global direction is removed, and group-specific residual directions.

#### 4.3 Activation and Steering Protocols

Activation extraction currently uses open-weight Qwen2.5 instruction models on Modal. Hidden states are mean-pooled over prompt tokens and saved with labels and metadata. Steering uses a signed direction NPZ, registers a forward hook at a selected transformer layer, adds a signed displacement during generation, and scores generated responses with the local rubric. Hidden-state telemetry measures whether the hook actually moves the targeted state before interpreting behavioral failure.

### 5. Results

#### 5.1 Cue-Balanced Boundary-Prior Activations

The controlled cue-balanced expansion removes the simple cue shortcut without breaking activation separation. The local scorer prefers contextual relation on 36/36 pairs with a mean score margin of +0.123 and autonomy-safety margin of +0.605. The simple cue counter solves 0/36 pairs, ties 36/36, and has mean cue margin 0.000.

| Model        | Layer | Dim  | Pairs | LOO Acc. | Mean Margin | Mean Cosine | Squared Acc. |
|--------------|-------|------|-------|----------|-------------|-------------|--------------|
| Qwen2.5 0.5B | -1    | 896  | 36    | 1.000    | +14.183     | +0.515      | 0.583        |
| Qwen2.5 0.5B | -2    | 896  | 36    | 1.000    | +2.732      | +0.371      | 0.639        |
| Qwen2.5 0.5B | -4    | 896  | 36    | 1.000    | +2.309      | +0.391      | 0.944        |
| Qwen2.5 1.5B | -1    | 1536 | 36    | 1.000    | +8.357      | +0.600      | 0.611        |
| Qwen2.5 1.5B | -2    | 1536 | 36    | 1.000    | +10.976     | +0.495      | 0.583        |

Table 1: Cue-balanced boundary-prior activation results. "Mean cosine" is the mean signed and absolute off-diagonal mechanism cosine; they match here because there are no high-absolute anti-aligned mechanism pairs.

#### 5.2 Geometry: Shared Manifold Plus Residual Mechanisms

The activation separation should not be described as independent orthogonal axes. Mechanism directions are moderately positively aligned. The stronger interpretation is a shared signed genuine-vs-pseudo manifold with fault-specific residual structure. Residual audits show that, after removing the global direction, mechanism-specific directions still separate all six groups on the boundary-prior expansion.

| Diagnostic                        | Primary-fault suite | Autonomy suite | Boundary-prior expansion |
|-----------------------------------|---------------------|----------------|--------------------------|
| Mean signed off-diagonal cosine   | +0.624              | +0.136         | +0.371 to +0.600         |
| Mean absolute off-diagonal cosine | 0.624               | 0.193          | 0.371 to 0.600           |

| Diagnostic                              | Primary-fault suite                  | Autonomy suite                   | Boundary-prior expansion                     |
|-----------------------------------------|--------------------------------------|----------------------------------|----------------------------------------------|
| Residual energy after global projection | 0.391                                | 0.828                            | Residual mechanism directions still separate |
| Responsible interpretation              | Shared manifold plus fault residuals | Less collapsed autonomy subspace | Shared signed boundary-prior manifold        |

Table 2: Geometry audits directly address the "orthogonality" and one-direction-ablation critiques. Signed reporting is required because squared projection energy erases the social pole.

### 5.3 Affect Control: Cohesion Is Not Just Positive Affect

Inspired by NOVA's EEG emotion-decoding emphasis on careful affect labels, PSD/DE feature choices, and simple ridge baselines, we added a text-only affect-control lane. This is not an EEG result. It crosses cue-balanced boundary-prior contrasts with six affect frames: anger, sadness, fear, disgust, happy, and neutral. The result is 72 matched pairs and 144 activation prompts.

| Control                        | Pairs | Accuracy | Mean Margin | Min Margin |
|--------------------------------|-------|----------|-------------|------------|
| Local scorer                   | 72    | 1.000    | +0.122      | +0.030     |
| Affect-only ridge LOO          | 72    | 0.750    | n/a         | n/a        |
| Affect-residualized scorer LOO | 72    | 1.000    | +0.116      | +0.017     |

Table 3: First affect-control residualization. Affect proxies explain some structure, but the local scorer's contextual-relation preference survives coarse affect/style residualization on this deterministic set.

### 5.4 Causal Steering: Hidden Movement Without Semantic Control

The first generation-time steering results are weak or mixed. Small generated-token post-hook edits reach a 0.750 prompt-level positive-vs-negative win rate, but the mean text-score shift is only +0.004. Stronger steering can separate generated outputs after re-embedding, while text scores move the wrong way. A dense -6 to +6 dose run shows that negative steering pushes generated outputs downward in projection, but positive steering does not lift them above baseline.

| Layer | Delta Error | Post-hook Pos-Baseline | Post-hook Pos-Neg | Text Score Shift |
|-------|-------------|------------------------|-------------------|------------------|
| -1    | 0.0073      | +5.2515                | +11.1520          | +0.0147          |
| -2    | 0.0018      | +5.9534                | +11.6887          | +0.0239          |
| -4    | 0.0025      | +5.7390                | +11.5992          | +0.0239          |

Table 4: Hidden-state telemetry. The hook works mechanically, but hidden displacement is not yet sufficient for semantic control.

## 6. Discussion

The most interesting result is not "we found the cohesion vector." The evidence points to a more publishable and more careful story: social constructs can be represented enough for strong probes on controlled examples, but the path from probe to causal control is nontrivial. This is a useful negative result because activation steering papers often blur classification, monitoring, and control.

The boundary-prior framing also gives the work a distinct angle. Many prosocial benchmarks reward agreement, politeness, or reduced conflict. Here, the positive class often keeps conflict alive: it protects dissent, verification, privacy, appeal, and exit. The model is not asked to dissolve boundaries. It is asked to use boundaries flexibly, contextually, and with consent.

The geometry audits substantially improve the paper's credibility. Reviewer critiques about near-orthogonality, sign-erasing squared projection, and single-direction ablations are right. The current result should be reported as a signed manifold plus residual subspaces, not as a set of independent axes or as proof that one vector exhausts the construct.

## 7. Limitations

These are synthetic compute-only results. The examples are deterministic or controlled wrappers, labels come from local rubrics, and the strongest results do not yet include independently generated paraphrases or human labels. No claim is made about human cooperation, persuasion, wellbeing, neural activity, fMRI, EEG, fNIRS, or hyperscanning. Any such claim requires IRB-appropriate human validation.

The local scorer remains a research instrument rather than ground truth. Metrics-only results can be circular when labels are derived from the same components. The next benchmark must reduce lexical and rubric leakage through independently generated paraphrases, adversarial wording, and pairwise human preference checks.

The steering result is not yet a success. It is a diagnostic. A future causal claim should require monotonic agreement among hidden-state telemetry, generated-output projection, pairwise behavior, and guardrail preservation.

## 8. Next Experiments

1. **Generated paraphrase hardening.** Replace controlled genre wrappers with generated/API-authored paraphrases, then rerun leakage, scorer, activation, geometry, residual, and signed-vs-squared reports.
2. **Affect residualization in activation space.** Extract activations for the affect-control prompt set, learn affect directions and cohesion directions in the same model/layer files, then test whether cohesion survives affect-direction regression.
3. **Monotonic steering protocol.** Sweep hook sites, generated-token-only edits, strength schedules, and decoding settings. Treat a setting as promising only if hook telemetry, output projection, and pairwise text behavior move together.
4. **Anti-compliance controls.** Add paired prompts where the cohesive response must refuse pressure, preserve evidence, or defend exit rights. A method that increases agreement while weakening these checks has failed.
5. **Human validation only after compute hardening.** Use a Prolific pairwise pilot once generated examples are no longer solved by lexical or rubric leakage. Neural work should wait for a robust behavioral signal.

## 9. Broader Impacts and Ethics

A system that optimizes "cohesion" can easily become a manipulation system. The project therefore treats truthfulness, autonomy, dissent, privacy, consent, appeal, and exit rights as constraints, not decorations. The benchmark intentionally includes warm-sounding harmful examples so that politeness, unity, and agreement cannot substitute for agency-preserving relation.

Human-facing experiments should measure manipulation risk and should not present model-generated social advice as validated intervention. Any study involving human participants requires appropriate review, consent, and risk disclosure.

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## NeurIPS-Style Checklist Summary

This compact checklist is included for share-readiness. A formal NeurIPS submission should use the official checklist bundled with the current LaTeX template.

| Item                                 | Answer  | Justification                                                                                                                               |
|--------------------------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Claims match evidence?               | Yes     | Claims are limited to compute-only benchmark, activation, geometry, and steering-telemetry results.                                         |
| Limitations disclosed?               | Yes     | Section 7 states synthetic data, rubric circularity, no human/neural claims, and failed steering caveats.                                   |
| Experimental details sufficient?     | Partial | The share draft summarizes methods. Reproduction details live in the repository scripts and experiment log.                                 |
| Error bars/statistical significance? | No      | Current deterministic smoke tests report margins and leave-one-pair-out results; generated/human validation should add intervals.           |
| Compute resources described?         | Partial | Experiments use local Python tooling and Modal GPU activation extraction on open-weight Qwen models.                                        |
| Ethics and societal impact?          | Yes     | Section 9 treats manipulation, autonomy, dissent, privacy, and human validation requirements as central risks.                              |
| Human subjects?                      | N/A     | No human participants were used for the reported results.                                                                                   |
| LLM/agent use?                       | Yes     | The experiments use open LLMs for activation extraction and generation-time steering. Codex assisted with writing/editing this share draft. |